

MATHEMATICAL METHODS IN SIGNAL PROCESSING

ECEN:	601-600	CRN:	11088
Lecture:	TR 9:35 AM – 10:50 AM	Location:	ETB 1003
Instructor:	Dr. Krishna Narayanan		krn@tamu.edu
	WEB 334K		

Course Description: The goal of this course is to develop within the student a degree of mathematical maturity. This maturity will be cultivated by working problems, following and constructing proofs, and writing programs. The main topics covered in this course include Hilbert spaces, orthogonal projections, representations and approximations, linear operators, pseudoinverses, matrix factorizations, characteristic values, singular value decomposition, and basic theory of constrained optimization. Applications of the mathematical tools presented in class to the fields of machine learning, signal processing and communications will be provided.

Learning Objectives

1. Establish basic notions of topology in the context of metric spaces. Study formal definitions for open sets, closed sets, convergence, limit points, completeness and continuous functions.
2. Review linear algebra, combinations of vectors, independence, bases and dimensions. Distinguish between vector spaces, normed spaces and inner-product spaces. Go over the projection theorem and illustrate some of its applications.
3. Introduce the notions of linear operators, fundamental subspaces, matrix representations, inverses and pseudoinverses. Examine the properties of characteristic polynomials, eigenvalues, eigenvectors and eigenfunctions. Develop the theory of the singular value decomposition. Survey special matrices and important matrix factorizations.
4. Apply vector space methods to signal processing, data science, optimization, least-squares filtering and minimum mean-square error estimation. Analyze iterative methods and the back propagation algorithm.
5. Acquire the ability to recognize, formulate and solve pertinent engineering problems using vector space methods and Hilbert spaces. Gain proficiency at using high-level programming languages such as C++ or Python.
6. Engage the student in an active learning experience. Expose the student to search engines, scholastic resources, research tools, indexes and databases. Prepare the student to become an active contributor to the common body of knowledge.

Text book: The course will be taught mainly out of two text books.

- **ECEN 601 Course Notes** by J.-F. Chamberland and H.D. Pfister
- **Mathematical Methods and Algorithms for Signal Processing** by T. K. Moon and W. C. Stirling

Recommended Texts: There exist several books that offer an excellent treatment of *Mathematical Methods in Signal Processing*.

- **Introduction to Applied Linear Algebra: Vectors, Matrices, and Least Squares** by S. Boyd and L. Vandenberghe
- **Linear Algebra** by K. Hoffman and R. Kunze
- **Optimization by Vector Space Methods** by D. G. Luenberger
- **Matrix Analysis** by R. A. Horn and C. R. Johnson

Exam Schedule (Dates may be slightly adjusted)

Test 1: October 3, 2024
Test 2: November 5, 2024
Final: December 5, 2024

Course Topics

Unit	Topics	Reading
1	Logic and Set Theory	Ch. 1
2	Metric Spaces and Topology	Ch. 2
3	Linear Algebra	Ch. 3
4	Representations and Approximations	Ch. 4
5	Linear Transformations and Operators	Ch. 5
6	Matrix Factorization and Analysis	Ch. 6
7	Canonical Forms	Ch. 7
8	Singular Value Decomposition	Ch. 8
9	Convexity, Lagrangian Optimization	

Grade Policies: The major grade components for *Mathematical Methods in Signal Processing* and their respective weights are listed below. Assignment and test grades will only be discussed after class or during office hours. We reserve the right to ask students to present their concerns or arguments in writing. Failure to meet a deadline may result in a grade of zero for the corresponding work.

Grading Rule

Assignments	30 %	Exam 1	20 %
		Exam 2	20 %
		Exam 3	30 %

The Academic Rules website at Texas A&M University, and its section on Grading in particular, discusses possible grades and their respective meaning:

<http://student-rules.tamu.edu/rule10>.

Academic Integrity Statement and Policy: “An Aggie does not lie, cheat or steal, or tolerate those who do.” For additional information, please visit: <http://aggiehonor.tamu.edu>.

Americans with Disabilities Act (ADA) Policy Statement: The Americans with Disabilities Act (ADA) is a federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation requires that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you believe you have a disability requiring an accommodation, please contact Disability Services, currently located in the Disability Services building at the Student Services at White Creek complex on west campus or call 979-845-1637. For additional information, visit <http://disability.tamu.edu>.

Helpful Links: Academic Calendar, Final Exam Schedule, Student Rules.